CH#13Uncertainty

Things don't happen as you expect
There's always a chance that

Contents

- Uncertainty
- Probability
- Syntax and Semantics
- Inference
- Independence and Bayes' Rule

Uncertainty

The cost of choosing

I need to be at the airport at 9:00 pm

- Agent 1: 95% chance of making it departing at 7:00pm
- Agent 2: 98% chance of making it departing at 8:00pm
- Agent 3: 100% chance of making it departing at 10:00pm

Which one do you choose? Why?

Agent 2: 98% chance of making it departing at 8:00pm

Uncertainty
Example

What would need to happen in the KB to diagnose a cavity.

- $Toothache \Rightarrow Cavity$
- $Toothache \Rightarrow Cavity \lor GumProblem \lor Abscess \ldots$
- $Cavity \Rightarrow Toothache$

But these can’t be true or complete...

Degree of Belief

Probabilities are a way to summarize uncertainty

- With the facts I can observe ...
- There's 80% of chance of having a cavity

Decision Theory

Likelihood and Importance

Rational decisions involve:

- Likelihood of achieving a state/goal (probability)
- Relative importance of the states/goals (Utility)

Decision Theory = probability theory + utility theory

Maximum Expected Utility: An agent is rational iff it chooses the action that yields the highest expected utility, averaged over all the possible outcomes of the action.

Probabilities in AI

Arguments Against[2]

- Prob. require massive amount of data
- Use of prob. require enumeration of all possibilities
- Hides details of character of uncertainty
- People are bad prob. estimators
- We do not have those numbers
- We find their use inconvenient

[2] Taken from V. Lesser's CS683

Introducing Probabilities in AI

Probability= It is a measure of how likely the event is to happen

It lies between 0-1

Probability

Likelihood Line

It will Snow in winter

The answer to 2 + 7 = 9

A year when it does not rain

Toss a coin That land Heads

Going without Food and drink for a year.

Probabilities

language of propositions

- **Random Variable:** variables like die_1, *cavity* , *etc.*
- **Domain:** The set of possible values for a variable. e.g. $die_1 = \{1, 2, 3, 4, 5, 6\}$

What's the domain of *Weather* , *Cavity* , *Age*[3]

Probabilities

- **A *random variable* can take on one of a set of different values, each with an associated probability. Its value at a particular time is *subject to random variation*.**
  - *Discrete* random variables take on one of a discrete (often finite) range of values
  - Domain values must be *exhaustive* and *mutually exclusive*
- **For us, random variables will have a discrete, countable (usually finite) domain of *arbitrary values*.**
  - Mathematical statistics usually calls these *random elements*
  - **Example: Weather is a discrete random variable with domain {sunny, rain, cloudy, snow}.**
  - ***Example: A Boolean random variable* has the domain {true,false},**

Let's consider $Weather = \{sunny, rainy, cloudy, snow\}$

- $P(sunny) = 0.6$; $P(rain) = 0.1$; $P(cludy) = 0.29$; $P(snow) = 0.01$
- $P(Weather)$ is a probability distribution.
- $P(Weather) =$

sunny	rainy	cloudy	snow
0.6	0.1	0.29	0.01

- $P(Weather) = < 0.6, 0.1, 0.29, 0.01 >$ for short

Probabilities
Concepts

- **Sample Space**: All possible worlds
  - dice roll: $\Omega = (1, 6)$
  - Ω = all possible worlds; ω = one world
- $0 \leq P(\omega) \leq 1$ for every ω
- $\text{P}_{\omega \in \Omega} P(\omega) = 1m$

Probability =No: of Favorable events/Total no; of events

What's the possiblity of obtaining a head in a coin toss?
1/2
Possibility of odd numbers in Die?

e.g., P(1)=P(2)=P(3)=P(4)=P(5)=P(6)=1/6.

Probabilities

More concepts and one example

- **Unconditional/Prior Probabilities:** $P(\textit{dice are even})$ or $P(\textit{doubles})$
- **Evidence:** Observed events. e.g. first die is 3.
- **Conditional/Posterior probabily:** given priors and evidence we compute the posterior.
  - e.g. Prob of doubles given that first die is 3.
  - $P(\textit{doubles} \mid \textit{die}_1 = 3)$
  - $P(\textit{cavity} \mid \textit{toothache} = T)$

Joint probability distribution

- **Probability assignment to all combinations of values of random variables (i.e. all elementary events)**

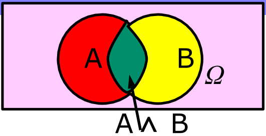
	toothache	¬ toothache
cavity	0.04	0.06
¬ cavity	0.01	0.89

- **The sum of the entries in this table has to be 1**
- ***Every question about a domain can be answered by the joint distribution***

- **Probability of a proposition is the sum of the probabilities of elementary events in which it holds**
  - P(cavity) = 0.1 [marginal of row 1]
  - P(toothache) = 0.05 [marginal of toothache column]

Conditional Probability

	toothache	¬ toothache
cavity	0.04	0.06
¬ cavity	0.01	0.89



- *P(cavity)=0.1* and *P(cavity ∧ toothache)=0.04* are both *prior* (unconditional) probabilities
- Once the agent has new evidence concerning a ***previously unknown*** random variable, e.g. Toothache, we can specify *a posterior* (conditional) probability e.g. ***P(cavity | Toothache=true)***

$$P(a \mid b) = P(a \wedge b)/P(b)$$

Where $P(b) > 0$

[*Probability of a with the Universe Ω restricted to b*]

→ ***The new information restricts the set of possible worlds ω_i consistent with it, so changes the probability.***

Conditional Probability (continued)

- **Definition of Conditional Probability:**

  $P(a \mid b) = P(a \wedge b)/P(b)$

- **Product rule gives an alternative formulation:**

  $$\begin{aligned} P(a \wedge b) &= P(a \mid b) * P(b) \\ &= P(b \mid a) * P(a) \end{aligned}$$

- **A general version holds for whole distributions:**

  $\mathbf{P}(Weather, Cavity) = \mathbf{P}(Weather \mid Cavity) * \mathbf{P}(Cavity)$

Independent Events - With Replacement.

- $P(A \cap B) = P(A)\,P(B/A) = P(B)\,P(A/B)$
- $P(A/B) = P(A)$ $\quad\quad$ $P(B/A) = P(B)$
- $P(A \cap B) = P(A)\,P(B)$ Independent.

2Red 3Black → One by one (R B)

$P(RB) = \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$ Ans (Without)

$P(R\,B) = \frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$ (With Replacement)

$P(RB) + P(BR) = \frac{2}{5} \times \frac{3}{5} + \frac{3}{5} \times \frac{2}{5} = \frac{12}{25}$ Ans

Probabilistic Inference

- ***Probabilistic inference:*** **the computation**
  - from *observed evidence*
  - of *posterior probabilities*
  - for *query propositions*.
- **We use the *full joint distribution* as the "knowledge base" from which answers to questions may be derived.**
- **Ex: three Boolean variables *Toothache (T), Cavity (C), ShowsOnXRay (X)***

	t		¬t	
	x	¬x	x	¬x
c	0.108	0.012	0.072	0.008
¬c	0.016	0.064	0.144	0.576

- **Probabilities in joint distribution sum to 1**

Probabilistic Inference II

	t		¬t	
	x	¬x	x	¬ x
c	0.108	0.012	0.072	0.008
¬c	0.016	0.064	0.144	0.576

- **Probability of any proposition computed by finding atomic events where proposition is true and adding their probabilities**
  - ***P(cavity V toothache)***
    = 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064
    = 0.28
  - ***P(cavity)***
    = 0.108 + 0.012 + 0.072 + 0.008
    = 0.2
- ***P(cavity)*** **is called a** ***marginal probability*** **and the process of computing this is called** ***marginalization***

Probabilistic Inference III

	t		¬t	
	x	¬x	x	¬ x
c	0.108	0.012	0.072	0.008
¬c	0.016	0.064	0.144	0.576

- **Can also compute conditional probabilities.**
- ***P(¬ cavity | toothache)***

  = P(¬ cavity ∧ toothache)/P(toothache)

  = (0.016 + 0.064) / (0.108 + 0.012 + 0.016 + 0.064)

  = 0.4

- **Denominator is viewed as a** *normalization constant***:**
  - Stays constant no matter what the value of Cavity is.
    (Book uses α to denote normalization constant 1/P(X), for random variable X.)

Bayes' Rule & Diagnosis

Likelihood

Prior

Posterior

$$P(a|b) = \frac{P(b|a) * P(a)}{P(b)}$$

Normalization

- **Useful for assessing diagnostic probability from causal probability:**

$$P(Cause|Effect) = \frac{P(Effect|Cause) * P(Cause)}{P(Effect)}$$

Bayes' Rule For Diagnosis II

$$P(Disease \mid Symptom) = \frac{P(Symptom \mid Disease) * P(Disease)}{P(Symptom)}$$

Imagine:

- **disease = TB, symptom = coughing**
- ***P(disease | symptom)* is different in TB-indicated country vs. USA**
- ***P(symptom | disease)* should be the same**
  - It is more widely useful to learn *P(symptom | disease)*

- **What about P(symptom)?**
  - Use *conditioning* (next slide)
  - For determining, e.g., the *most likely* disease given the symptom, we can just ignore P(symptom)!!! (see slide 35)

Conditioning

- *Idea:* **Use *conditional probabilities* instead of joint probabilities**
- $P(a) = P(a \wedge b) + P(a \wedge \neg b)$

$= P(a \mid b) * P(b) + P(a \mid \neg b) * P(\neg b)$

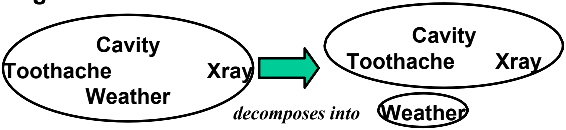
Here:

$P(symptom) = P(symptom \mid disease) * P(disease) + P(symptom \mid \neg disease) * P(\neg disease)$

- **More generally: $P(Y) = \sum_z P(Y|z) * P(z)$**
- **Marginalization and conditioning are useful rules for derivations involving probability expressions.**

The Solution: *Independence*

- **Random variables A and B are *independent* iff**
  - ***$P(A \land B) = P(A) * P(B)$***
  - ***equivalently: $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$***
- ***A and B are independent if knowing whether A occurred gives no information about B (and vice versa)***
- **Independence assumptions are *essential* for efficient probabilistic reasoning**



$$P(T, X, C, W) = P(T, X, C) * P(W)$$

- **15 entries (2^4-1) reduced to 8 (2^3-1 + 2-1)**
  For *n independent* biased coins, *$O(2^n)$ entries →$O(n)$*

Conditional Independence

- **BUT *absolute* independence is rare**
- **Dentistry is a large field with hundreds of variables, none of which are independent. What to do?**

- **A and B are *conditionally independent given C* iff**
  - $P(A \mid B, C) = P(A \mid C)$
  - $P(B \mid A, C) = P(B \mid C)$
  - $P(A \wedge B \mid C) = P(A \mid C) * P(B \mid C)$

- **Toothache (T), Spot in Xray (X), Cavity (C)**
  - None of these are independent of the other two
  - But ***T and X are conditionally independent given C***